



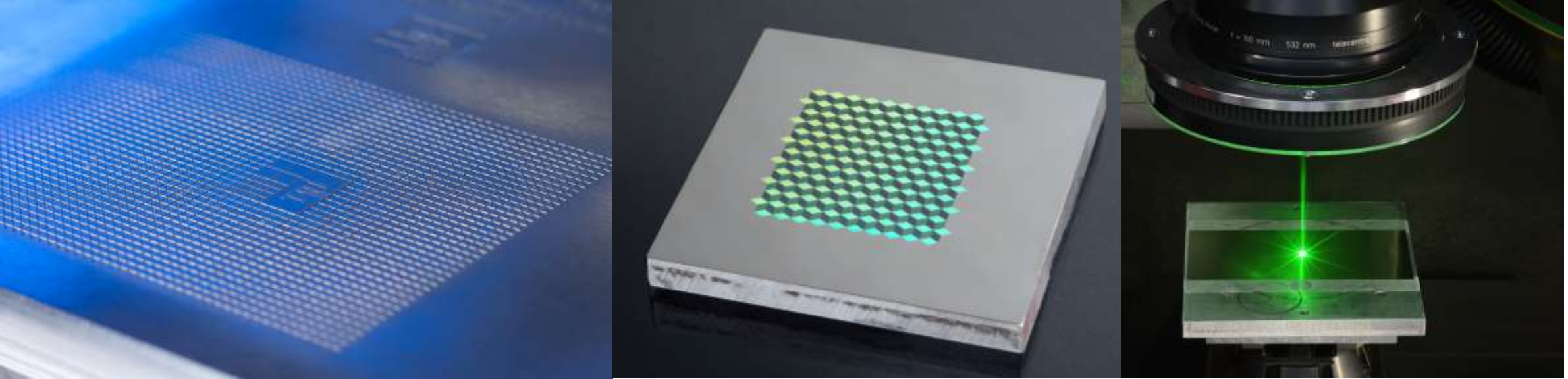
Sahil Nair
M.Sc. Mechatronics
FB 8 – Mechanical Engineering and Mechatronics
Fachhochschule Aachen



Prof. Dr.-Ing. Kristian Arntz
Professor & Dean
FB 8 – Mechanical Engineering and Mechatronics
Fachhochschule Aachen



M.Sc. Duc Thanh Tran
Research Fellow Directed Energy Manufacturing
Fraunhofer Institute for Production Technology IPT



Master's Thesis Colloquium

A Python-Based Toolpath Generation Framework for Robot-Based Laser Structuring

Internal supervisor: Prof. Dr.-Ing. Kristian Arntz

External supervisor: M.Sc. Duc Thanh Tran

Sahil Nair | 3552894 | M.Sc. Mechatronics

TABLE OF CONTENTS

Introduction

Methodology

CAD Segmentation

Robot Path Planning

Laser Path Planning

Validation

Final Results & Discussion

Conclusion

Future Work

INTRODUCTION



MOTIVATION & RESEARCH CONTEXT



Laser structuring: Enables surface functionalization



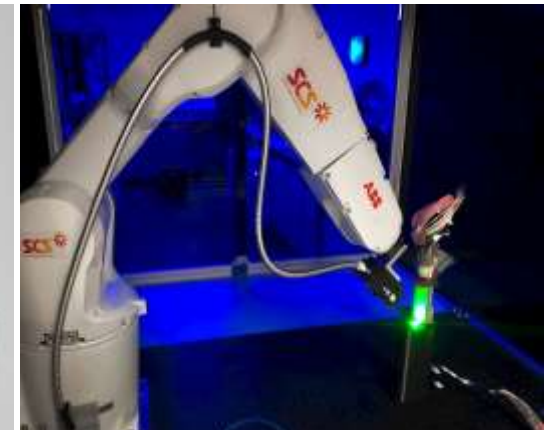
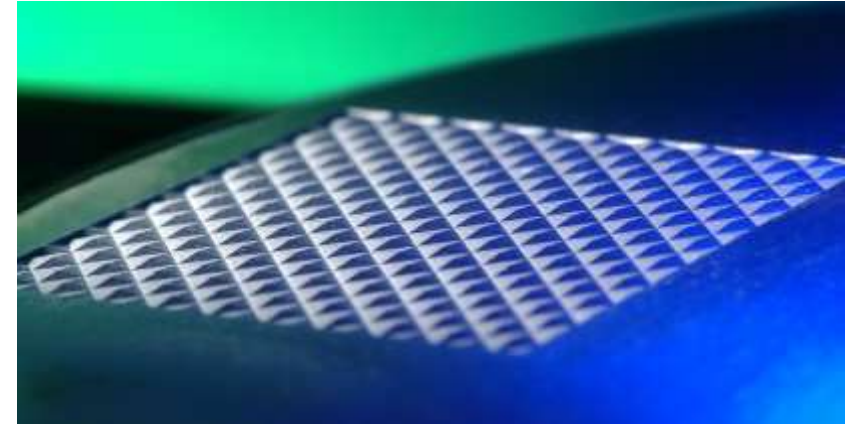
Industrial Relevance: Applications in aerospace, automotive, biomedical devices, and optics



Challenge: Large, curved 3D surfaces are difficult to process with traditional fixed systems



Research Gap: Robot-based laser systems allow flexibility but need segmentation + optimized toolpath planning



PROBLEM STATEMENT & CHALLENGES



Toolpath generation is essential in modern manufacturing



Large, curved 3D surfaces are complex to laser structure



Current C++ based toolpath generation framework (CALM)

Large legacy C++ codebase
Lacks proper documentation
Absence of unit tests



This results in:

Frequent crashes
Uninformative error messages
Black-box nature limits troubleshooting and innovation
Difficulty in maintenance and upgrading

THESIS OBJECTIVES



Create a **Python-based pipeline** to automate toolpath generation for laser structuring of large and curved surfaces



Leverage open-source tools, frameworks, and libraries in building the software solution



Validate the developed solution with existing hardware to assess its integration at Fraunhofer IPT

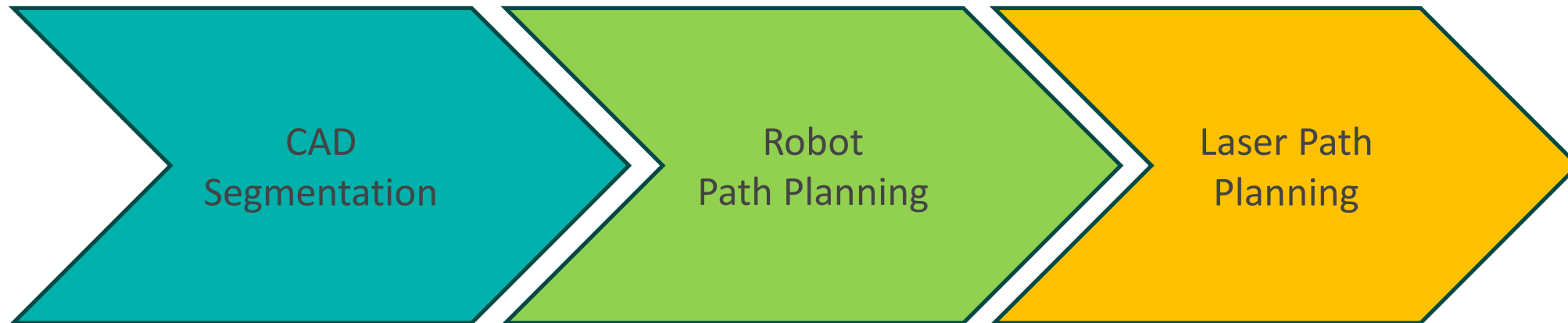


METHODOLOGY



METHODOLOGY

- To overcome the limitations of the legacy CALM system, a modular, Python-based workflow was developed consisting of three interconnected stages



TOOLS & LIBRARIES

1. CAD Segmentation

PyVista
Mamba

python

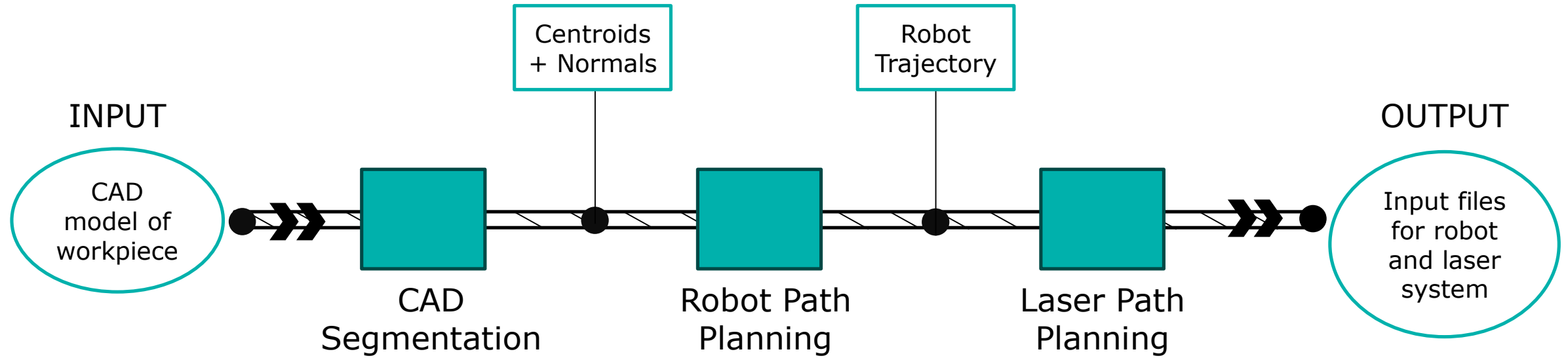
2. Robot Path Planning

ROS2
MoveIt2

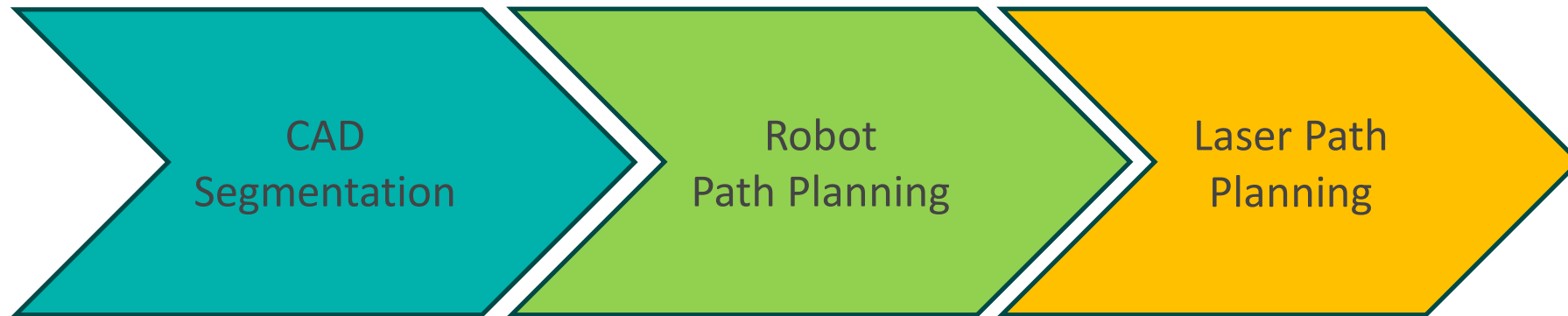
3. Laser Path Planning

FreeCAD

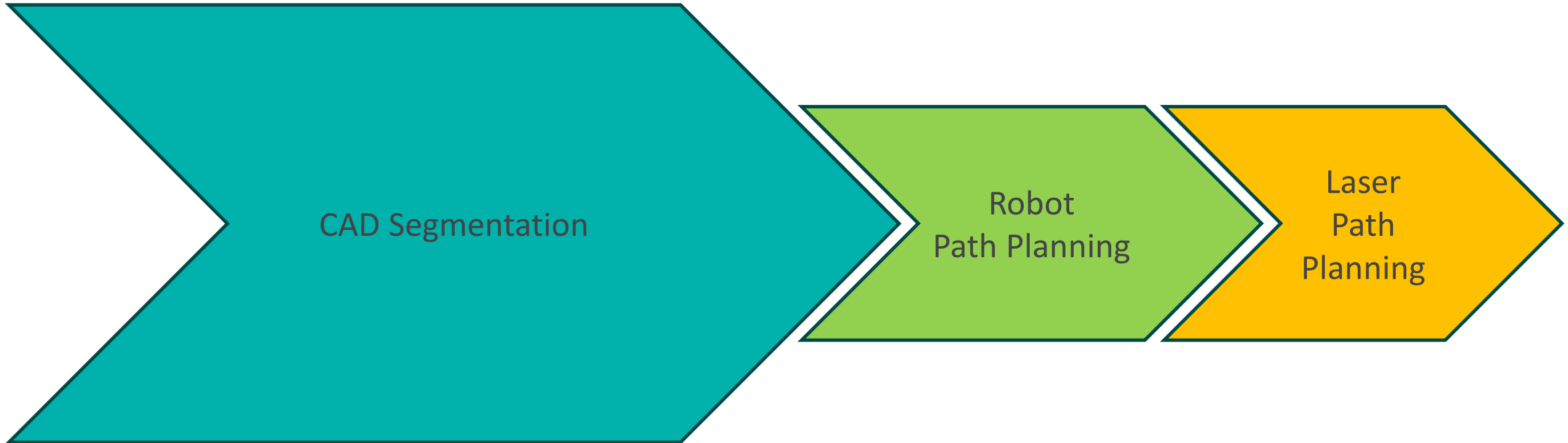
SYSTEM ARCHITECTURE



3 STAGE PIPELINE

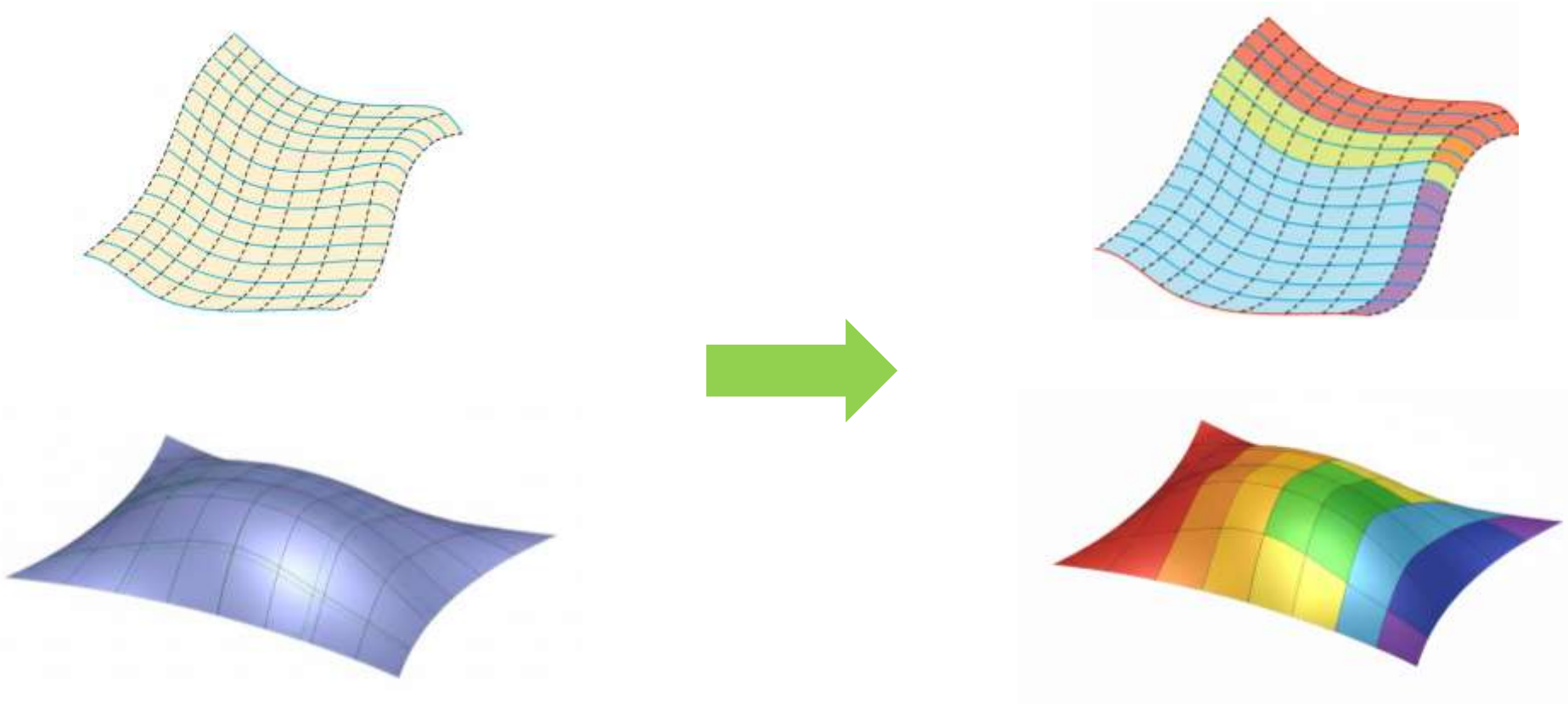


FIRST STAGE



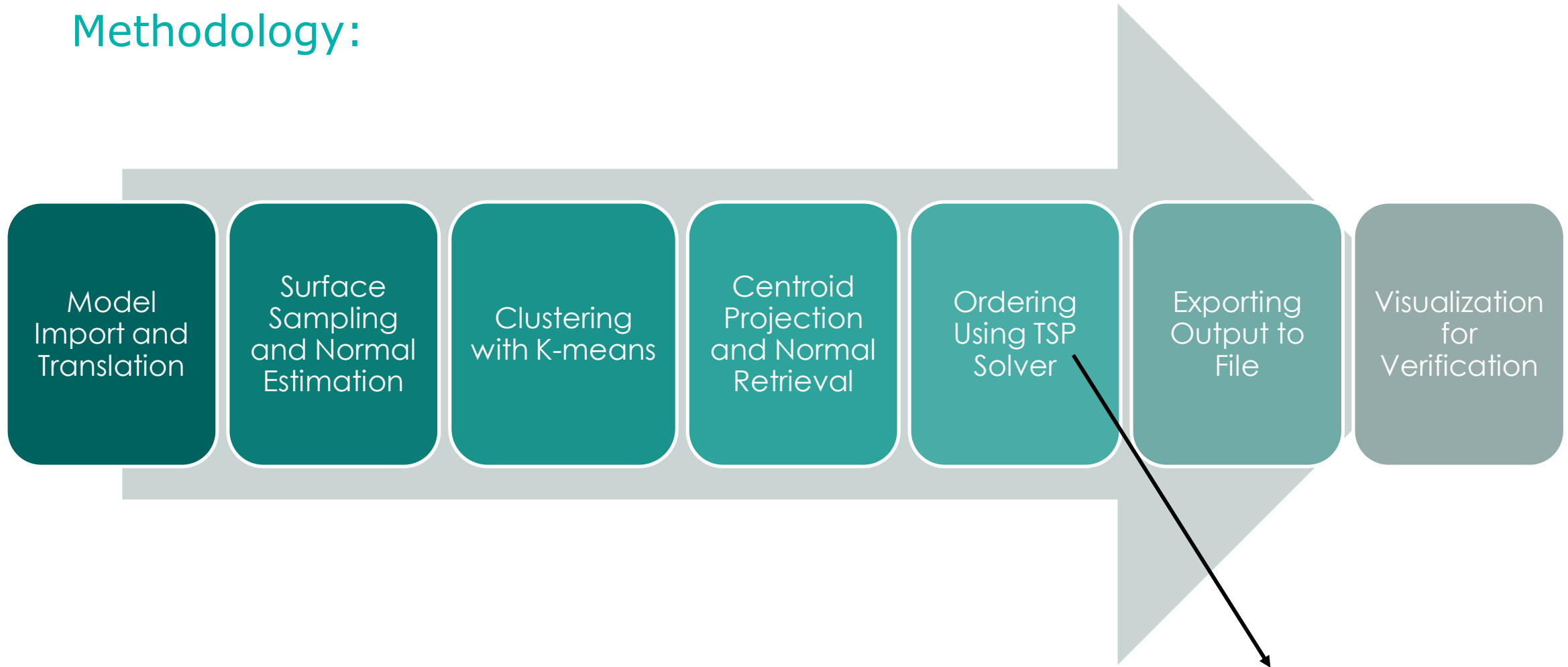
CAD SEGMENTATION

Goal: Divide complex CAD surfaces into smaller, processable regions (segments or clusters) for laser operations



CAD SEGMENTATION

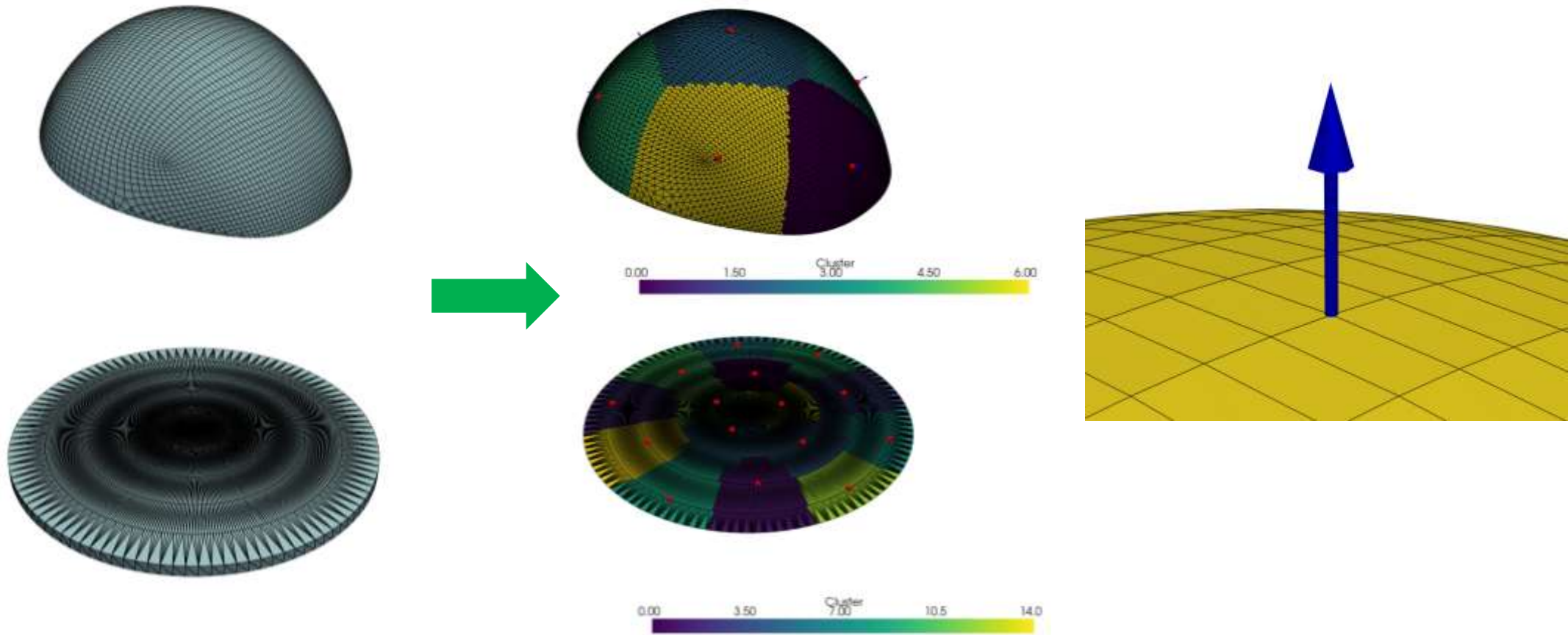
Methodology:



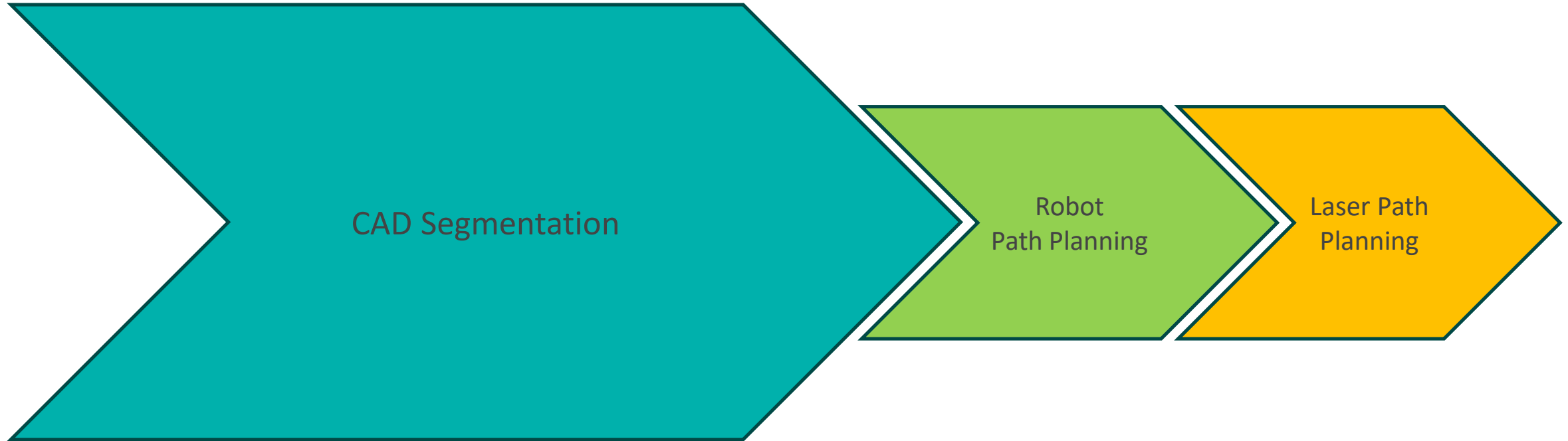
TSP = Travelling Salesman Problem

CAD SEGMENTATION

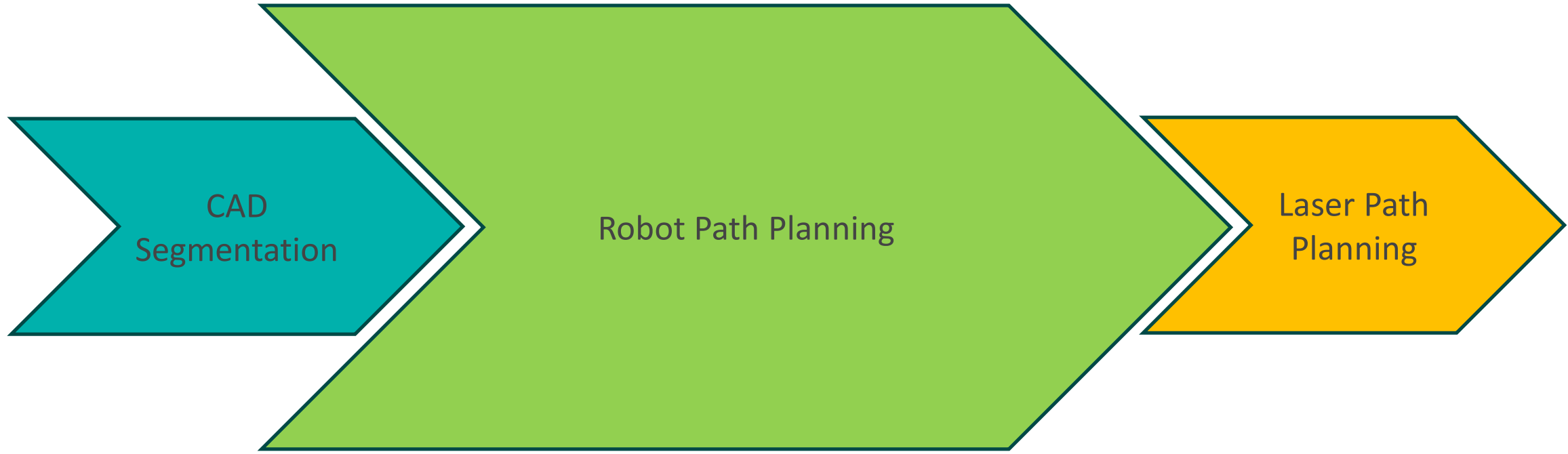
Results: visuals showing segmentation before/after, centroid points overlaid on model



FIRST STAGE

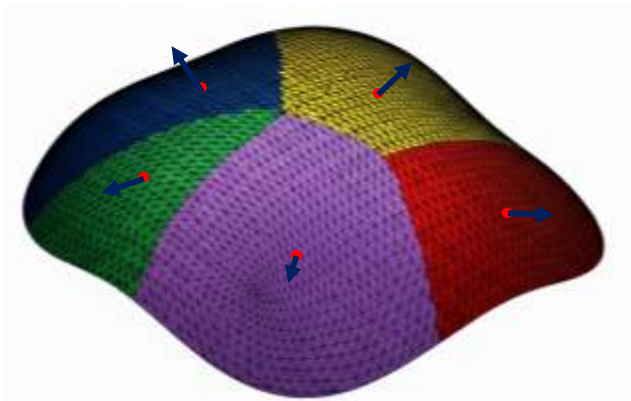


SECOND STAGE

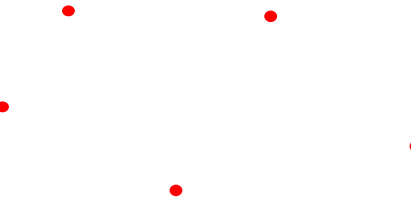


ROBOT PATH PLANNING

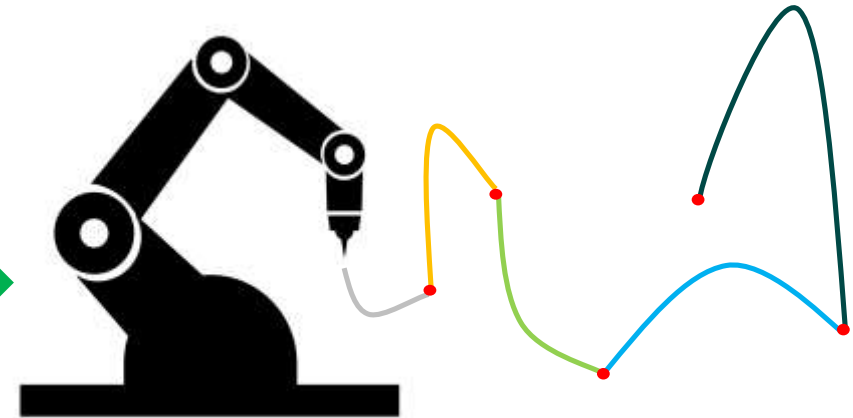
Goal: To compute feasible **collision-free joint trajectories** that accurately position the robot's tool **perpendicular to the local surface normal** at each segmented region.



Segmented object



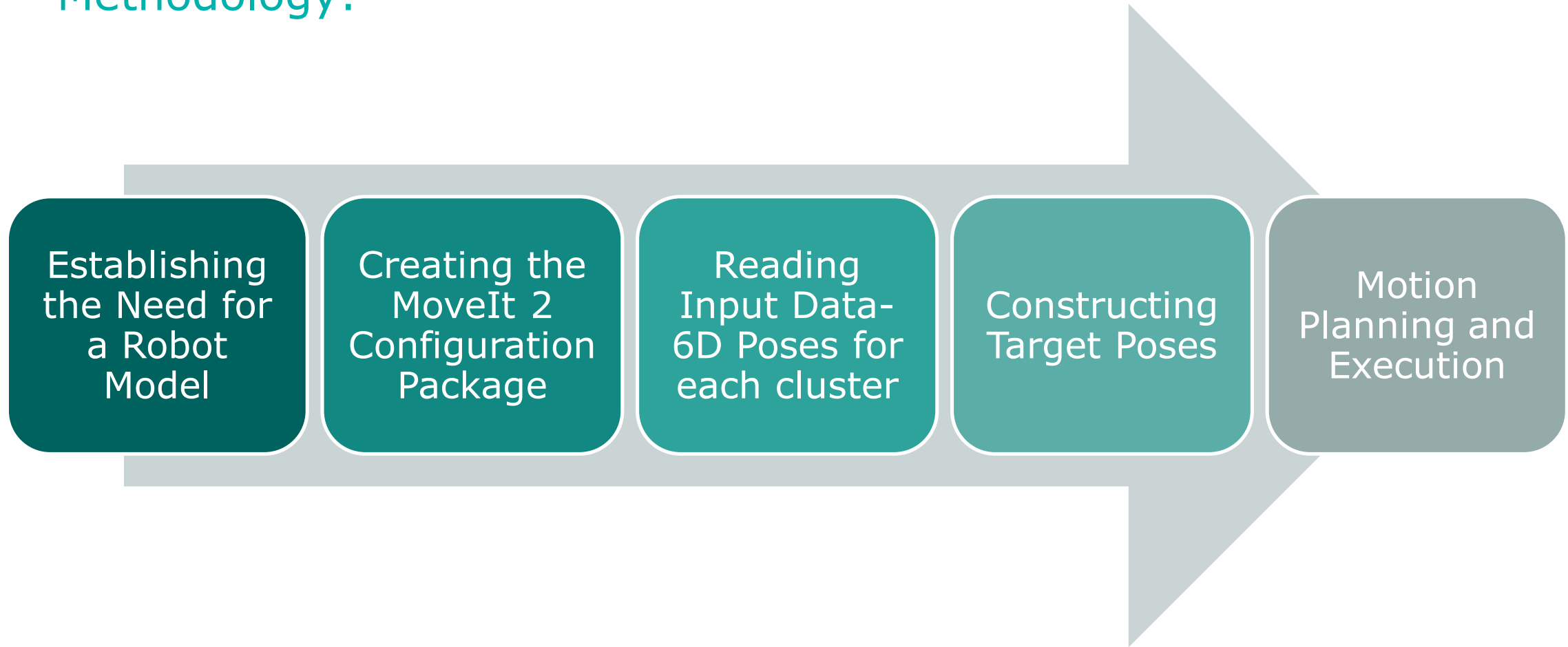
Poses
(Position + Orientation)



Trajectory

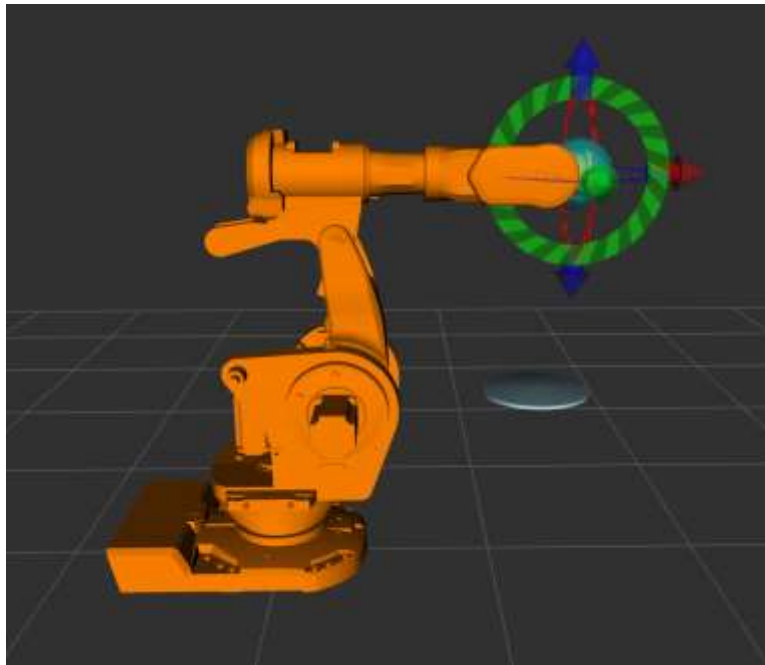
ROBOT PATH PLANNING

Methodology:

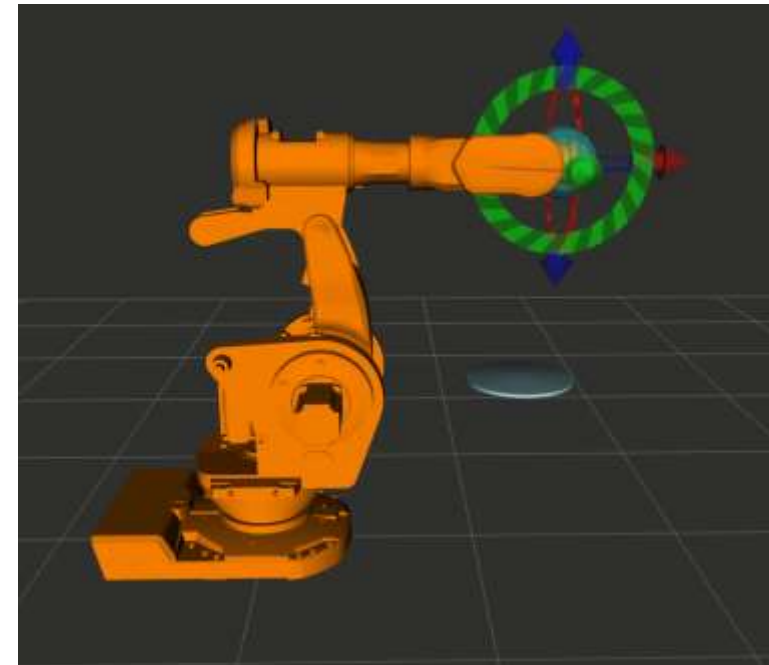


ROBOT PATH PLANNING

Results: visuals showing ABB robot's URDF model and motion planning with MoveIt 2 visualised in RViz

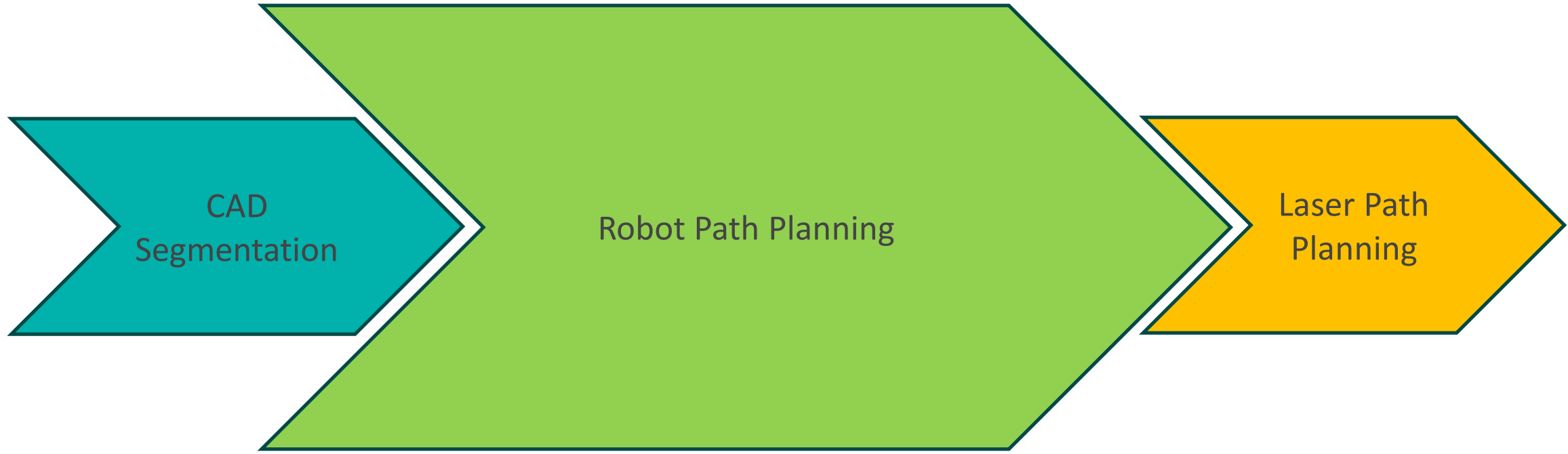


- Object(blue): Segmented workpiece

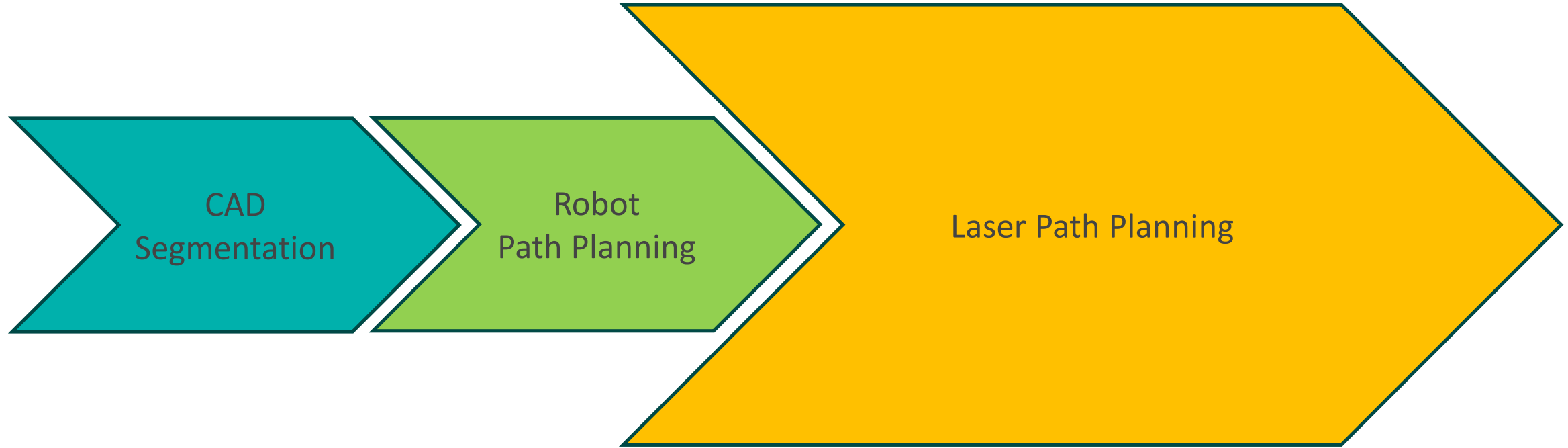


- Robot executing planned trajectory
- Robot arm's end aligned perpendicular to the surface

SECOND STAGE

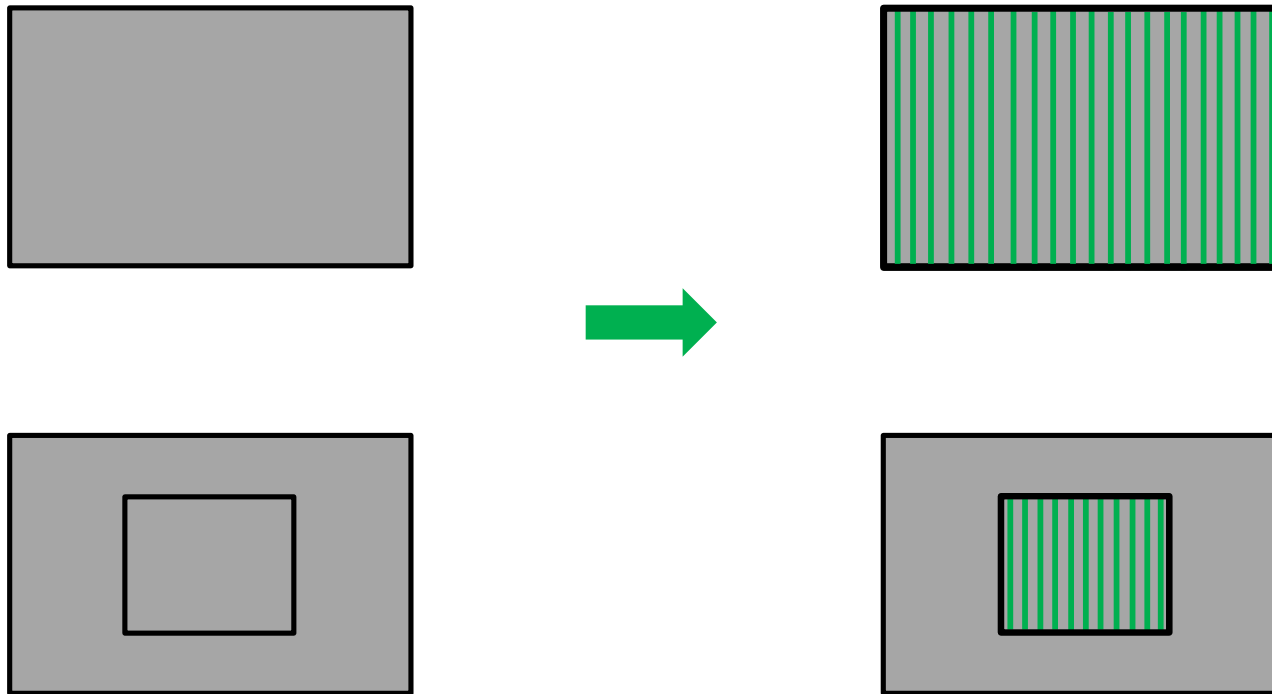


THIRD STAGE



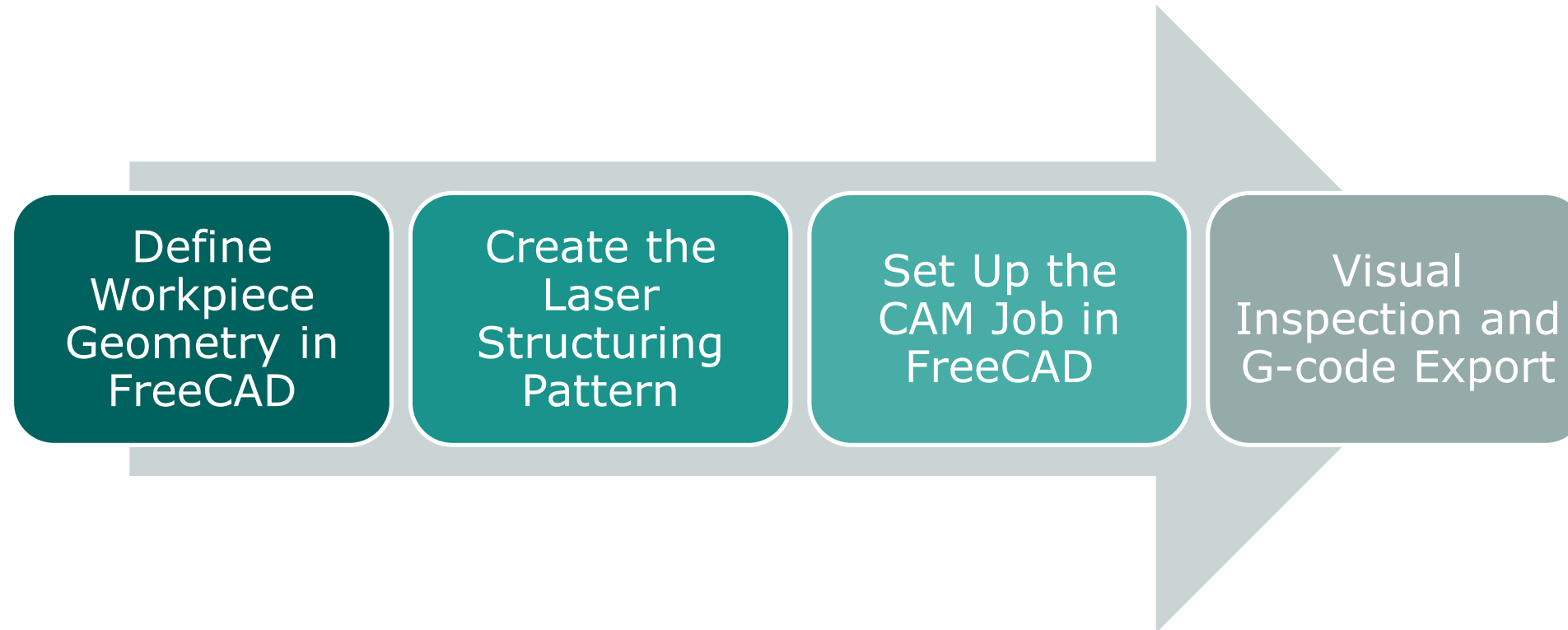
LASER PATH PLANNING

Goal: To **generate structured scan patterns (G-code)** on simplified, segmented surface patches to ensure **uniformity and coverage** for the laser-material interaction.



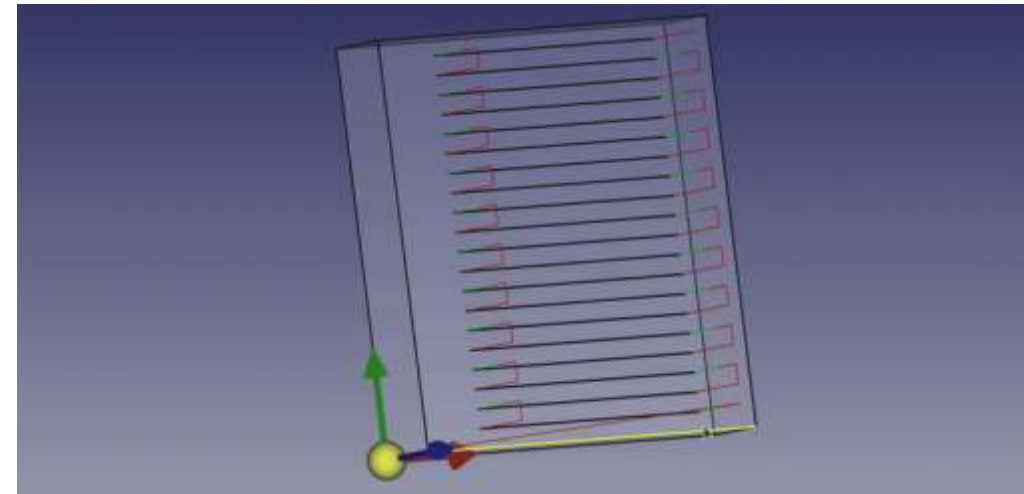
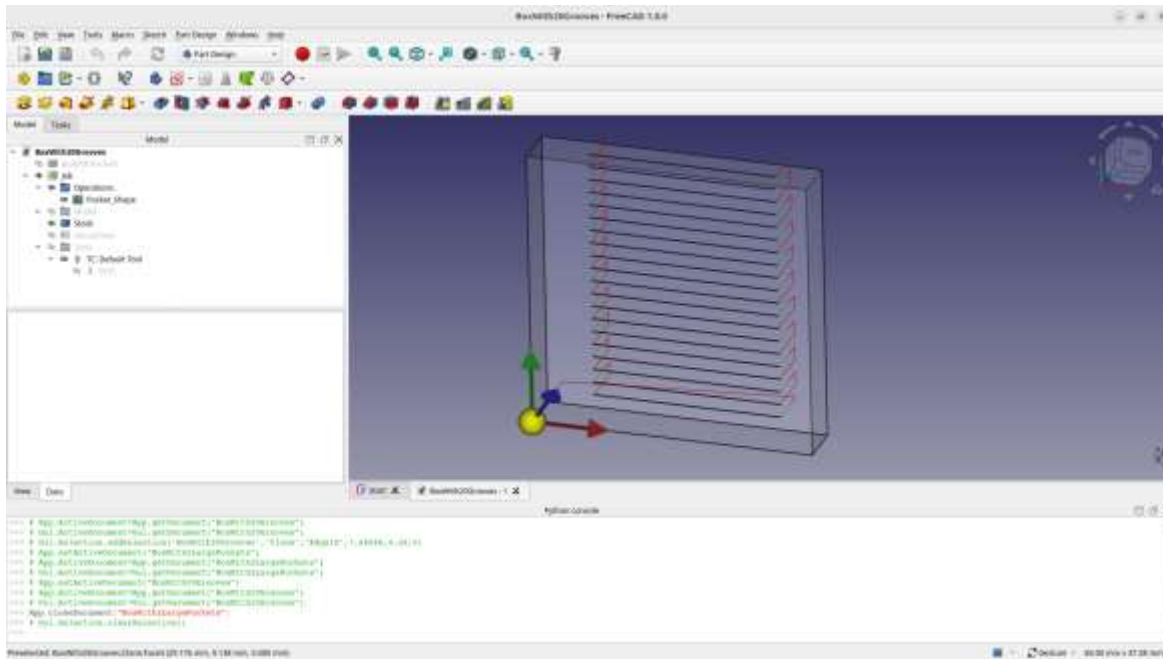
LASER PATH PLANNING

Methodology:



LASER PATH PLANNING

Results: Simulation screenshots of robot executing the toolpath, Table of metrics: success rate, computation time, coverage



VALIDATION



VALIDATION

Goal: To ensure each stage functions and produces outputs aligning with practical execution workflow at Fraunhofer IPT

Strategy: Involved multiple levels of assessment

- Geometric and Numerical Checks
- System Integration and Compatibility
- Physical Output Validation
- Post-Processing Development

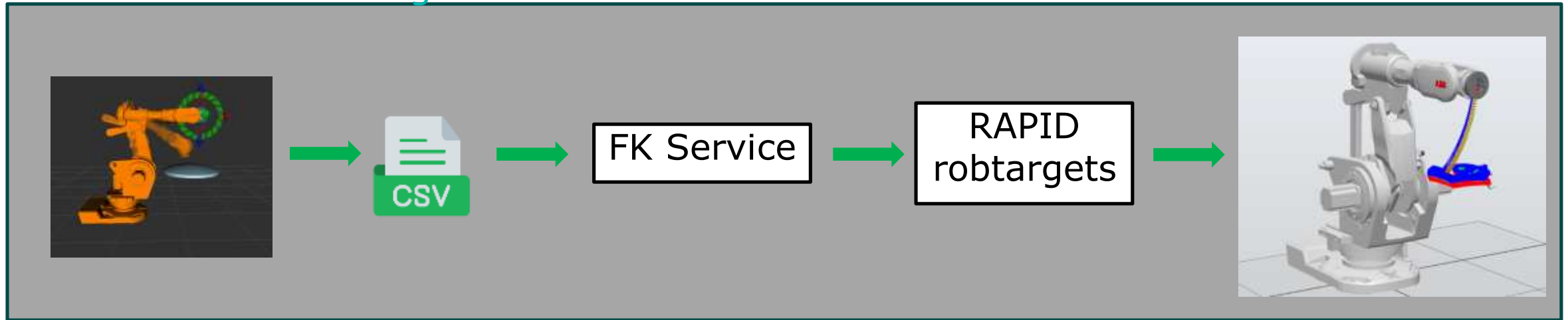
CAD Segmentation



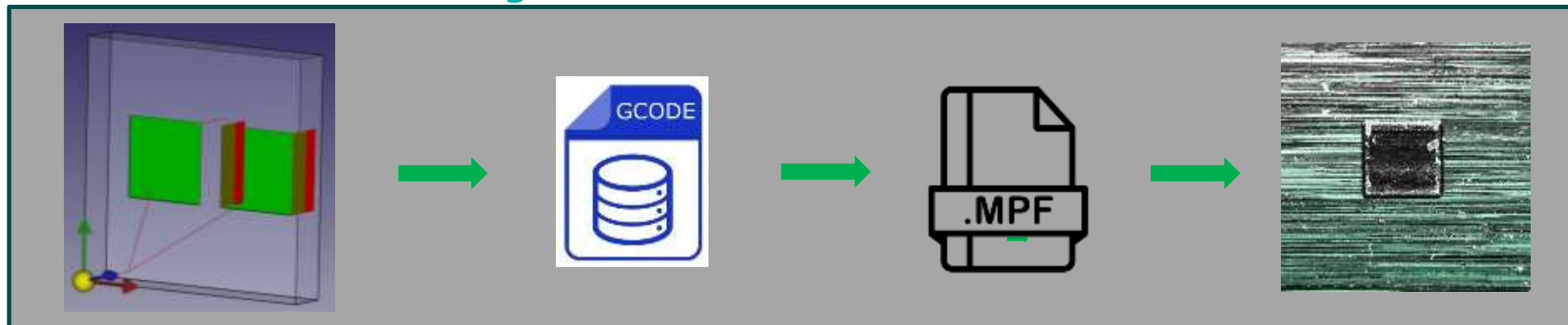
$$\text{Coverage (\%)} = \frac{\sum \text{Area of Segments}}{\text{Total Surface Area of Mesh}} \times 100$$

Cluster size < Laser window size

Robot Path Planning



Laser Path Planning

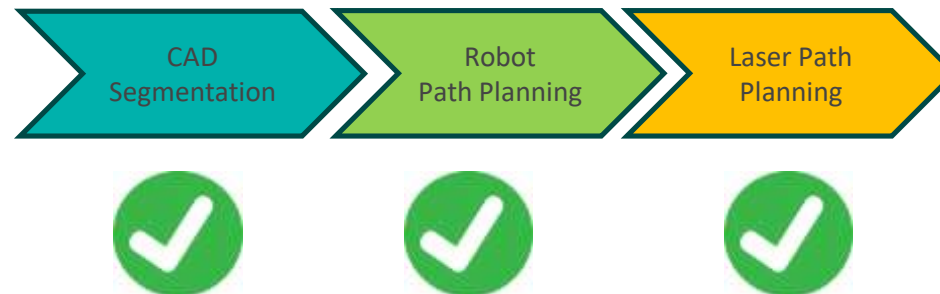


FINAL RESULTS & DISCUSSION



FINAL RESULTS

- **CAD Segmentation:** Successfully clustered complex surfaces into planar regions using **PyVista**, validated through visual inspection and surface coverage
- **Robot Path Planning:** Generated accurate, collision-free joint trajectories (MoveIt 2), validated in **ABB RobotStudio**, confirming reachability
- **Laser Path Planning:** Produced valid G-code paths using **FreeCAD**, validated through physical laser ablation tests



TECHNICAL CONSIDERATIONS AND CHALLENGES

1. Cluster Count Estimation

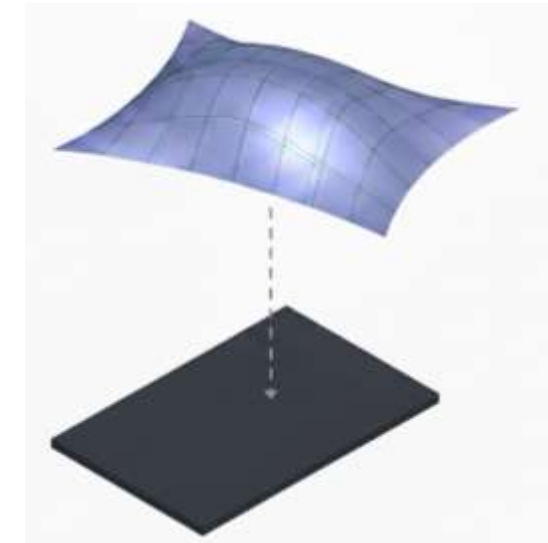
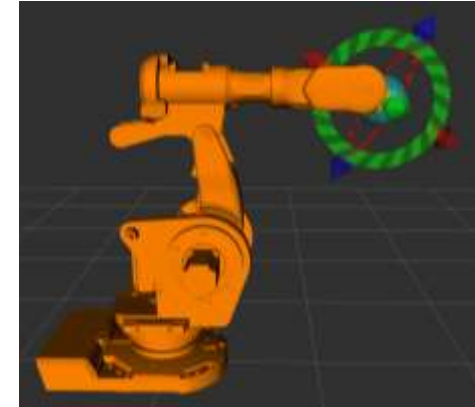
- Manually determined based on a practical estimation

2. URDF Limitations with ABB IRB-6660-205/1.9

- Limitation of URDF prevented inclusion of support links D1 & D2

3. Limited Surface Complexity

- Flat planar surfaces were used for laser structuring as a practical simplification

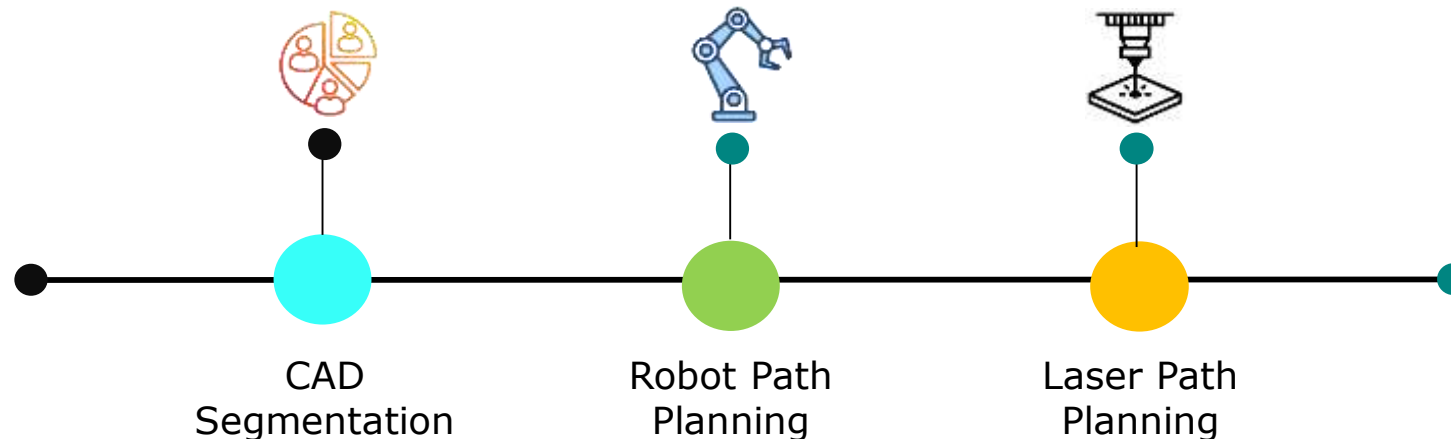


CONCLUSION



CONCLUSION

- Developed a **modular, python**-based workflow enabling robotic laser structuring on large, complex 3D surfaces.
- Combined CAD segmentation, robot path planning, and laser path planning using **open-source** tools.
- Achieved **end-to-end integration** and **physical validation** on Fraunhofer IPT's setup



FUTURE WORK



FUTURE WORK



Advanced
Visualization and
Simulation

Containerization
of Individual
Stages



Integration of
External Linear
Axis in Motion
Planning



ACKNOWLEDGEMENTS & QUESTIONS

I would like to express my sincere gratitude to:

- **Fraunhofer IPT** - providing the facilities, resources, and technical environment that made this work possible
- **M.Sc. Duc Thanh Tran** - for valuable guidance and encouragement
- **Prof. Dr.-Ing. Kristian Arntz** - for continuous support and supervision
- **My colleagues** - for discussions and technical input



APPENDIX



FUTURE WORK



State Machine
for Segment
Transition
Management

Automation of
CAM Operations

Toolpath
Optimization



Extending the
Framework to
Other
Applications

Generalization to
Complex, Real-World
Surfaces



REFERENCES

Industrial & Research Context

- <https://www.ilt.fraunhofer.de/en/technology-focus/surface-technology.html>
- <https://new.abb.com/news/detail/61133/it-is-rocket-science>
- [pyvista logo sm.png \(960×400\)](#)
- [freecad-logo.png \(3000×800\)](#)
- [Python-Logo.png \(3840×2160\)](#)
- [logo.png \(557×357\)](#)
- [ROS-2 logo.png \(1946×564\)](#)
- [moveit-2-logo-dark.png \(2544×1079\)](#)
- https://www.google.com/search?sxsrf=AE3TifPLsbMOJ1YhmfeNOMUmV0pJoAIcGQ:1760306285874&udm=2&q=cad+segmentation+icon#vhid=Dt4gQUvU_OCYqM&vssid=mosaic
- <https://www.google.com/search?q=robot+arm+icon&udm=2&sxsrf=AE3TifMmeSjpcO6XdCcCL-DwVAXuWH4qA%3A1760306506243#vhid=O4IfgwQ9eeJY3M&vssid=mosaic>
- <https://www.google.com/search?q=laser+structuring+icon&udm=2&sxsrf=AE3TifNIO8x1DSleNh4KnfAIzbdUx7xwNg%3A1760306651234#vhid=Vlvch1Llv2tmyM&vssid=mosaic>

CAD SEGMENTATION

Input



3D surface mesh

+

Optional parameters
(K, export path)



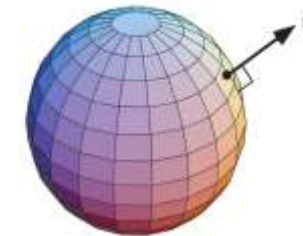
Output

CSV file consisting of :

- Projected Centroids



- Surface Normals



ROBOT PATH PLANNING

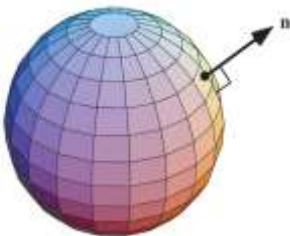
Input



Projected Centroids

+

Surface normals



Output

CSV file consisting of RobotTrajectory object

- sequence of joint configurations
- Later validated for collision-free execution



LASER PATH PLANNING

Input



Desired structuring pattern

+

Laser processing parameters



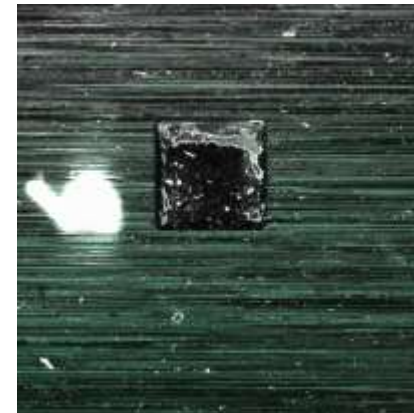
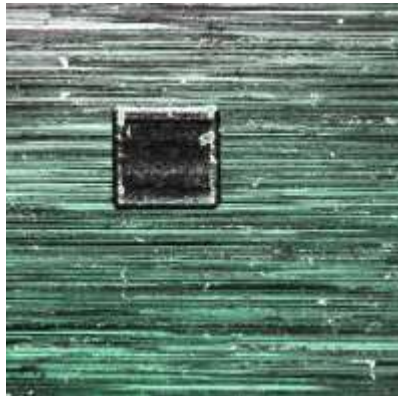
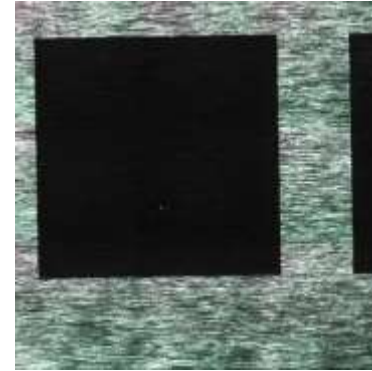
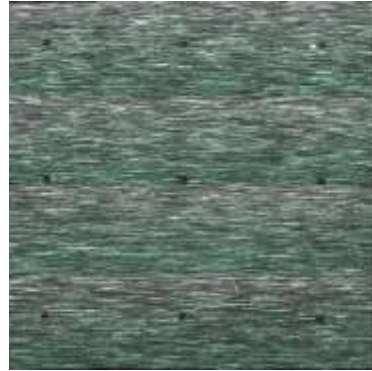
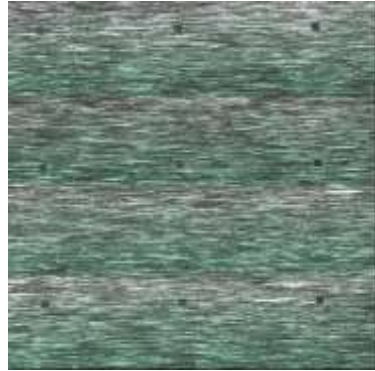
Output



G-code file



LASER STRUCTURING RESULTS



K-means Algorithm

- Segmentation process required grouping surface points into nearly planar clusters
- The method needed to be **scalable, geometry-agnostic, and computationally efficient** for large meshes

Rationale for selection

- **Simplicity & Efficiency:** Performs well on large 3D point datasets with limited computational overhead
- **Control Over Number of Segments (k):** Enables flexible adjustment of segmentation resolution.
- **Compatibility:** Easily implemented using **Python**



Pyvista

- Native support for both structured and unstructured 3D surface meshes
- Seamless integration with VTK[44]
- Intuitive and Pythonic API, well-suited for exploratory workflows
- Support for computing surface normals, extracting regions, and exporting processed data

Library	CAD Import	Mesh Processing	Surface Normals	Segmentation	Visualization	Maintenance Status	Notes
Pyvista	Via mesh	Yes	Yes	Yes	Yes	Active	Chosen tool
CadQuery	Yes	No	Limited	No	Limited	Active	Best for CAD generation
Trimesh	Via mesh	Yes	Yes	Manual	Limited	Active	Lightweight & fast
PyMesh	Yes	Yes	Yes	Limited	No	Deprecated	Difficult to integrate
Open3D	Yes	Yes	Yes	Yes	Yes	Active	Overhead if only surface segmentation needed

Robot Path Planning Tools

MoveIt 2

- open-source robotic manipulation framework designed for motion planning, kinematics, collision checking, and trajectory generation

Reasons

- Integrated Planning Pipeline
- Plugin-Based Architecture
- Visualization Support
- Python Integration



Robot Path Planning Tools

ROS 2

- underlying middleware supporting MoveIt 2 and the overall robotic system

Reasons

- Modern Middleware
- Scalability and Modularity
- Python Support
- Active Development and Community



Technical Considerations and Challenges- CAD Segmentation

1. Algorithm Selection and Simplification
2. Feature Vector Design
3. Cluster Count Estimation and Adaptability
4. Centroid Projection and Normal Alignment
5. Ordering via TSP Solver
6. Visualization and Debugging Support

Technical Considerations and Challenges- Robot Path Planning

1. ROS 2 Installation Strategy and Build Overhead
2. Compatibility of moveitpy with ROS 2 Versions
3. Visualization and Integration with RViz
4. URDF Limitations with ABB IRB-6660-205/1.9
5. Coordinate System Alignment and Pose Accuracy
6. Output Format and Pipeline Integration

Technical Considerations and Challenges- Laser Path Planning

1. Tool Representation and CAM Interpretation
2. Toolpath Strategy Adaptation
3. Limited Surface Complexity
4. Manual Setup of CAM Operations
5. Export Format and Post-Processing